

CAPM-based portfolio optimisation

Rolling estimation, constrained allocation and risk evaluation

A walk-forward comparison of CAPM-implied and historical-mean portfolio allocations, with trading constraints, transaction costs and explicit analysis of downside risk.

Methods	Python; rolling CAPM; Ledoit–Wolf covariance; constrained optimisation
Data	US sector ETFs and SPY; January 2007–December 2024
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1 Executive summary

This project compares constrained mean–variance portfolios across eleven US sector ETFs. The central design choice replaces historical mean–return estimates with rolling CAPM betas and a fixed 5.5% annual equity risk premium. Ledoit–Wolf covariance shrinkage, monthly rebalancing, position limits and transaction costs complete the allocation framework.

The base run produces 178 monthly return observations from March 2010 to December 2024. CAPM-MV has a reported annualised return-to-volatility ratio of 0.739, compared with 0.819 for historical-mean Naive MV and 0.881 for SPY. The charts call this measure “Sharpe”; it is calculated without subtracting the risk-free rate.

The comparison reveals a trade-off. CAPM-MV has a less severe maximum drawdown than Naive MV (−25.7% versus −30.7%), but higher historical one-month 95% Expected Shortfall (9.99% versus 9.18%). It leads Naive MV on the reported ratio in 2020–2024 and trails it in the earlier sub-periods.

A 24-month lookback gives a higher reported ratio of 0.897 in the sensitivity analysis. That is a retrospective parameter comparison, not evidence of a validated improvement. Synthetic pre-launch ETF histories and return-compounding conventions also limit how directly the reported backtest can be interpreted as an investable strategy.

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2 Data and scope

2.1 Objective and benchmarks

Mean–variance optimisation can be sensitive to estimated expected returns [1, 8]. This project tests whether a CAPM-based return structure produces a useful allocation trade-off relative to historical sample means. It imposes a modelling assumption on cross-sectional expected returns; it is not presented as an independently validated alpha signal.

Three benchmarks provide context:

- **SPY:** the passive market benchmark and CAPM market proxy.
- **Equal Weight (EW):** a monthly rebalanced allocation of 1/11 to each sector ETF.
- **Naive MV:** historical mean returns with the same covariance estimator and portfolio constraints as CAPM-MV.

The comparison with Naive MV focuses on the expected-return input. Its value is assessed across return, drawdown, tail risk and trading costs rather than through a single performance measure.

2.2 Universe and source

Monthly adjusted closing prices are sourced from Yahoo Finance for the period January 2007–December 2024. The three years prior to the backtest start date provide the initial rolling estimation window. The full universe is listed in Table 1.

Table 1. Investment universe and approximate S&P 500 sector weights (2024).

Ticker	Sector	Approx. SPY Weight
XLK	Technology	~29%
XLF	Financials	~13%
XLV	Health Care	~12%
XLY	Consumer Discretionary	~11%
XLI	Industrials	~9%
XLC	Communication Services	~9%
XLP	Consumer Staples	~6%
XLE	Energy	~4%
XLU	Utilities	~2%
XLRE	Real Estate	~2%
XLB	Materials	~2%
SPY	S&P 500 (benchmark / market proxy)	—

2.3 Return calculation

Asset returns are calculated as monthly log returns:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right). \quad (1)$$

Here $P_{i,t}$ is the adjusted month-end closing price. The 13-week US T-bill yield ($\text{\textasciitilde IRX}$) from Yahoo Finance is converted to a monthly decimal rate, and the estimation input is $r_{i,t}^e = r_{i,t} - r_{f,t}$.

2.4 Data quality

XLC was launched in June 2018 and XLRE in October 2015. Before launch, the implementation creates synthetic prices by multiplying SPY prices by the ratio of the ETF's launch price to SPY's price on that date. The constant scaling cancels in log returns, so wholly backfilled windows inherit SPY returns and a beta of 1.0. These histories are proxies, not returns on tradable pre-launch ETFs.

Adjusted closing prices account for splits and dividends. The completed analysis series has no material gaps in the backtest window, but synthetic observations remain an important limitation.

3 Methodology

3.1 Rolling CAPM expected returns

For each asset i and rebalance date t , a rolling OLS regression is estimated over a 36-month lookback window:

$$r_{i,s}^e = \alpha_i + \beta_{i,t} r_{m,s}^e + \varepsilon_{i,s}, \quad s = t - 36, \dots, t - 1, \quad (2)$$

where $r_{m,s}^e$ is the excess return on SPY at month s . The estimated beta $\hat{\beta}_{i,t}$ captures the asset's current systematic exposure to the market. Expected returns are then constructed as:

$$\mu_{i,t} = r_{f,t} + \hat{\beta}_{i,t} \cdot \frac{\text{ERP}}{12}, \quad (3)$$

where the equity risk premium ERP is set to 5.5% per annum and converted to monthly units, consistent with long-run US estimates [5].

The vector $\boldsymbol{\mu}_t = (\mu_{1,t}, \dots, \mu_{11,t})^\top$ is a model-implied return input. Its cross-sectional differences come from beta estimates and the fixed ERP. This reduces reliance on separately estimated sample means, while introducing CAPM and ERP assumptions. The backtest does not isolate estimation-error reduction as the cause of any performance difference.

Beta estimates are winsorised at the 5th/95th percentile of the cross-sectional distribution each month to prevent outlier betas from distorting the return vector.

3.2 Ledoit–Wolf covariance shrinkage

The sample covariance $\mathbf{S}_t$ computed from 36 months of returns is prone to overfitting: extreme eigenvalues are inflated, producing ill-conditioned inverses that destabilise the optimiser. Ledoit and Wolf [4] propose an analytical shrinkage estimator that convexly combines the sample covariance with a structured target $\mathbf{T}$:

$$\hat{\Sigma}_t = (1 - \delta) \mathbf{S}_t + \delta \mathbf{T}, \quad (4)$$

where $\delta \in [0, 1]$ is an analytically estimated shrinkage intensity. The supplied scikit-learn `LedoitWolf` implementation uses $\mathbf{T} = \bar{v} \mathbf{I}$, with $\bar{v} = \text{tr}(\mathbf{S}_t)/11$: an identity target scaled by average variance, rather than a constant-correlation target [9]. Shrinkage regularises the covariance estimate used by the optimiser.

3.3 Constrained mean–variance optimisation

At each rebalance date, portfolio weights $\mathbf{w} = (w_1, \dots, w_{11})^\top$ are chosen to maximise risk-adjusted return subject to practical constraints:

$$\max_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\mu}_t - \frac{\lambda}{2} \mathbf{w}^\top \hat{\Sigma}_t \mathbf{w} - c \|\mathbf{w} - \mathbf{w}_{t-1}\|_1 \quad (5)$$

subject to:

$$\begin{aligned} \mathbf{w}^\top \mathbf{1} &= 1, & (\text{full investment}) \\ w_i &\geq 0.02, \quad \forall i, & (\text{min. 2\% per sector}) \\ w_i &\leq 0.40, \quad \forall i, & (\text{max. 40\% per sector}) \\ \|\mathbf{w} - \mathbf{w}_{t-1}\|_1 &\leq \tau. & (\text{turnover cap}) \end{aligned}$$

The turnover cap is $\tau = 0.30$ per monthly rebalance and the base risk-aversion parameter is $\lambda = 1.5$. Expected returns and covariance are both expressed in monthly units. The penalty coefficient $c = 0.0010$ represents 10 basis points per unit of L1 turnover; it also discourages large changes in portfolio weights.

Before optimisation each month, $\mathbf{w}_{t-1}$ is updated by drifting the prior weights using realised returns (*drift adjustment*), so the turnover constraint correctly penalises intentional rebalancing rather than passive weight drift:

$$\|\mathbf{w} - \mathbf{w}_{t-1}\|_1 = \sum_{i=1}^{11} |w_i - w_{i,t-1}|. \quad (6)$$

The quadratic programme is solved using SLSQP. Across the walk-forward run, the solver converged successfully on every rebalance solve.

3.4 Walk-forward backtesting

The rolling estimation windows end before the return being evaluated. This supports a walk-forward design, subject to the synthetic-history and implementation limitations discussed later. The structure is summarised in Table 2.

Table 2. Walk-forward backtest structure.

Phase	Period	Detail
Initialisation	Jan 2007 – Jan 2010	Burn-in: populate 36-month estimation window
Backtest period	Mar 2010 – Dec 2024	178 reported monthly return observations; synthetic histories included
Rebalance frequency	Monthly (end-of-month)	Weights set using only data up to rebalance date
Estimation window	Rolling 36 months	Fixed-length; avoids look-ahead in beta estimation
Cost model	10 bps per unit trade	Deducted from portfolio return on net weight change

At each month-end t : (1) update the rolling return window; (2) estimate $\hat{\beta}_{i,t}$ via OLS for each sector ETF; (3) compute μ_t ; (4) compute $\hat{\Sigma}_t$ via Ledoit–Wolf; (5) drift prior weights by realised returns; (6) solve the constrained quadratic programme; (7) record net trades and costs; (8) apply weights to compute the next month’s portfolio return.

3.5 Performance measures

Metrics are computed over the full backtest period and separately over sub-periods (2010–2014, 2015–2019, 2020–2024) to assess consistency across regimes.

3.5.1 Return-to-volatility ratio (reported Sharpe)

The reported Sharpe ratio is computed from monthly net portfolio log returns as:

$$\text{SR} = \frac{\bar{r}_p \times 12}{\sigma_p \sqrt{12}}, \quad (7)$$

where $\bar{r}_p$ is the mean monthly net portfolio return and σ_p is the monthly return volatility. This provides a consistent annualised return-to-volatility comparison across strategies. The risk-free rate is used in the CAPM expected return construction, while the reported performance ratios are computed directly from realised net strategy returns. Because no risk-free return is subtracted in this metric, it is a return-to-volatility ratio, equivalent to assuming a zero risk-free rate for the reported Sharpe measure. Existing chart labels are retained with this convention made explicit.

3.5.2 Maximum drawdown

Drawdown is measured using the negative drawdown convention:

$$\text{DD}_t = \frac{V_t - \max_{s \leq t} V_s}{\max_{s \leq t} V_s}, \quad (8)$$

where V_t is cumulative portfolio value at time t . Maximum drawdown is then:

$$\text{MDD} = \min_t \text{DD}_t. \quad (9)$$

This convention reports drawdowns as negative percentages. The supplied implementation constructs wealth with $\prod_t(1 + r_{p,t})$ despite log-return inputs; the resulting drawdowns are retained as reported diagnostics. A production rerun should first reconcile the simple-return and log-return conventions.

3.5.3 Tracking error and information ratio

$$\text{TE} = \sigma(r_{p,t} - r_{\text{SPY},t}) \times \sqrt{12}, \quad \text{IR} = \frac{R_p - R_{\text{SPY}}}{\text{TE}}. \quad (10)$$

3.5.4 Calmar ratio

$$\text{Calmar} = \frac{R_p}{|\text{MDD}|}. \quad (11)$$

3.5.5 Turnover

Monthly L1 turnover is:

$$\text{Turnover}_t = \sum_{i=1}^{11} |w_{i,t} - w_{i,t-1}^*|, \quad (12)$$

where $w_{i,t-1}^*$ is the drifted prior weight. Net-of-cost returns deduct 10 bps per unit of L1 turnover per month. This convention is consistent with the turnover cap and transaction cost penalty used in the optimiser.

3.5.6 Historical VaR and Expected Shortfall

Tail risk is measured using one-month historical Value at Risk (VaR) and Expected Shortfall (ES) at the 95% confidence level. Both are reported using the positive-loss convention:

$$\text{VaR}_{0.95} = -q_{0.05}(r_{p,t}), \quad (13)$$

where $q_{0.05}(r_{p,t})$ is the empirical 5th percentile of monthly portfolio returns. Expected Shortfall is the average loss conditional on being in the VaR tail:

$$\text{ES}_{0.95} = -\mathbb{E}[r_{p,t} \mid r_{p,t} \leq q_{0.05}(r_{p,t})]. \quad (14)$$

Because the backtest uses monthly log returns, the VaR and ES figures should be interpreted as historical one-month log-return tail-loss estimates.

4 Results and discussion

4.1 Summary performance

Table 3 reports the reported backtest performance across all four strategies, net of 10 bps transaction costs.

CAPM-MV has a reported ratio of 0.739, below SPY (0.881) and Naive MV (0.819) over the same full period. Its maximum drawdown of -25.7% is five percentage points less severe than Naive MV's -30.7% , and its Calmar ratio is higher. Reported annual L1 turnover is 4.1%. However,

Table 3. Reported backtest performance, March 2010 – December 2024 realised returns. Net of 10 bps costs. CAPM-MV column shown in bold.

Metric	CAPM-MV	Naive MV	Equal Wt.	SPY
Ann. Return (net)	11.47%	12.23%	11.16%	13.07%
Ann. Volatility	15.53%	14.93%	14.72%	14.84%
Sharpe Ratio	0.739	0.819	0.758	0.881
Max Drawdown	−25.7%	−30.7%	−24.6%	−25.6%
1M VaR 95%	7.17%	7.12%	6.05%	6.61%
1M ES 95%	9.99%	9.18%	9.51%	9.19%
Calmar Ratio	0.447	0.399	0.454	0.511
Ann. L1 Turnover	4.1%	—	≈0%	0%
IR vs SPY	−0.73	−0.21	−0.66	—

one-month 95% VaR is 7.17% and ES is 9.99%, so the drawdown advantage does not translate into uniformly lower monthly tail losses.

4.2 Cumulative returns and drawdown

Figure 1 shows the reported growth of \$1 across the four strategies. The paths are similar through 2018 and diverge more thereafter. CAPM-MV reaches approximately \$4.8, compared with \$6.0 for SPY. Its full-period maximum drawdown is less severe than Naive MV's, but slightly more severe than SPY's. The compounding convention described in the methodology applies to these wealth paths.

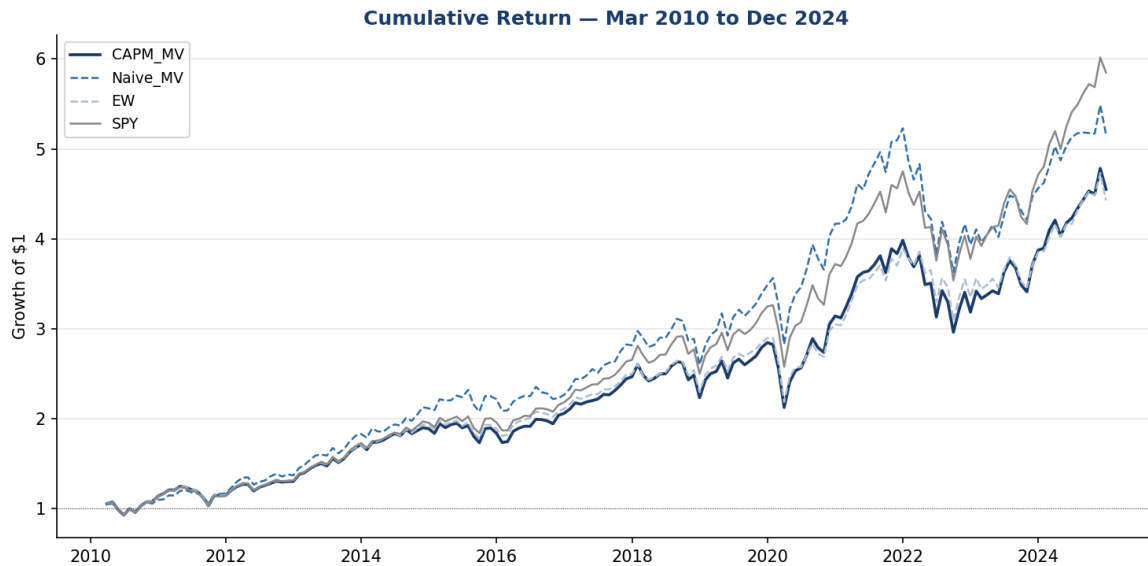


Figure 1. Cumulative return (growth of \$1), March 2010 – December 2024. All realised returns are net of 10 bps transaction costs.

Figure 2 plots drawdown through time with stress episodes shaded. The 2022 rate-shock episode (right shaded region) is particularly informative: CAPM-MV's drawdown is materially shallower than Naive MV, which reached -30.7% at its trough. The COVID crash (left shaded region) shows the opposite pattern—CAPM-MV's beta-structured tilt into cyclicals amplified the drawdown relative to SPY.

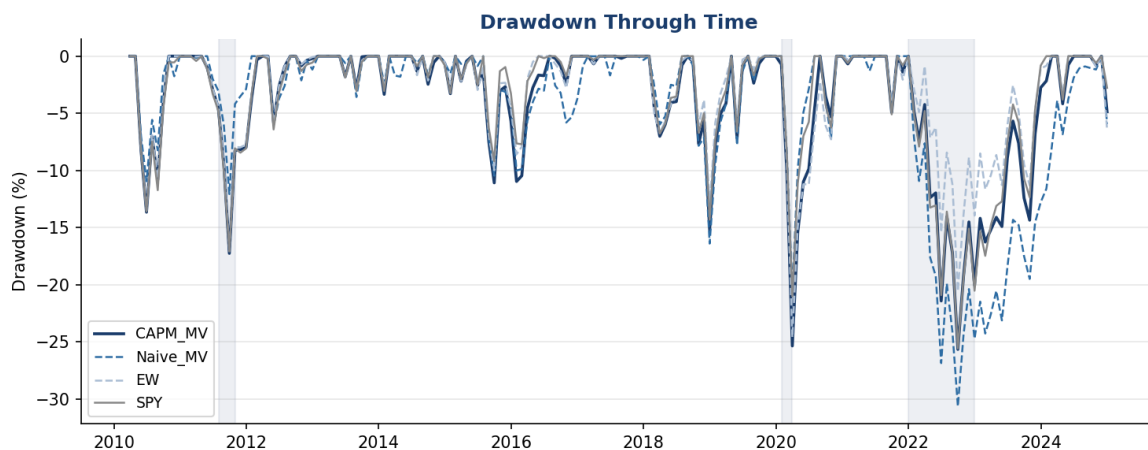


Figure 2. Drawdown through time. Shaded regions indicate the three stress scenarios analysed in Section 4.6.

4.3 Sector weight dynamics

Figure 3 shows quarterly snapshots of the CAPM-MV weights. XLK rises to around 21% by December 2024, while several sectors approach the 2% floor. The allocations reflect expected returns, covariance, bounds and turnover penalties jointly; the heatmap alone does not identify the contribution of each input.

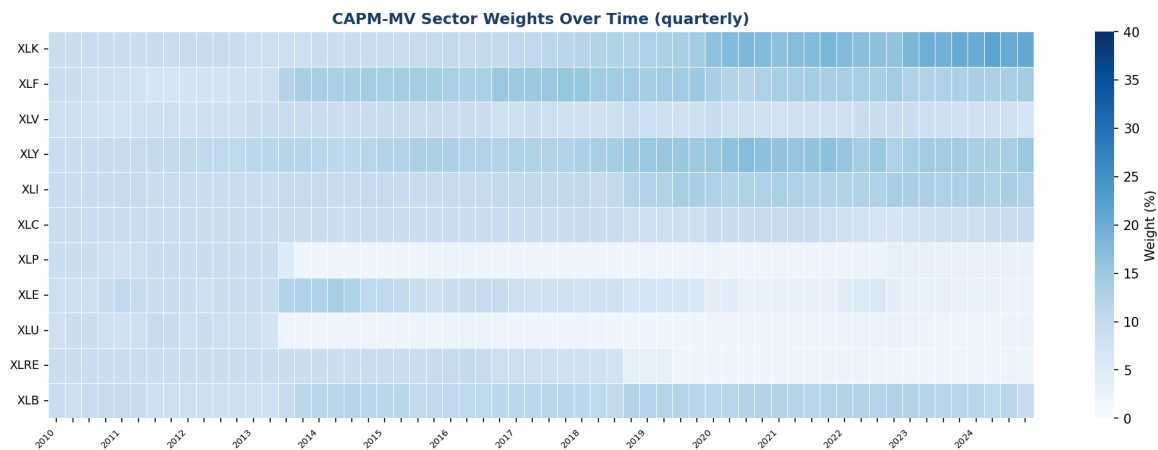


Figure 3. CAPM-MV sector weights over time (quarterly frequency). Darker shading indicates higher weight; white indicates near the 2% minimum.

4.4 Sub-period analysis

Table 4 and Figure 4 break performance down across three five-year sub-periods.

Table 4. Sub-period Sharpe ratios.

Period	CAPM-MV	Naive MV	Equal Wt.	SPY
2010–2014	1.078	1.360	1.110	1.124
2015–2019	0.704	0.832	0.772	0.895
2020–2024	0.575	0.511	0.547	0.731

The ranking changes across sub-periods. CAPM-MV trails Naive MV by approximately 0.28 ratio points in 2010–2014 and 0.13 in 2015–2019, then leads by approximately 0.06 in 2020–2024. This is consistent with a regime-dependent trade-off, but does not by itself establish that differences in mean-estimation error caused the reversal.

Figure 5 plots the rolling 12-month reported Sharpe measure. The strategies alternate in relative performance, reinforcing the importance of examining sub-periods rather than relying only on full-sample averages.

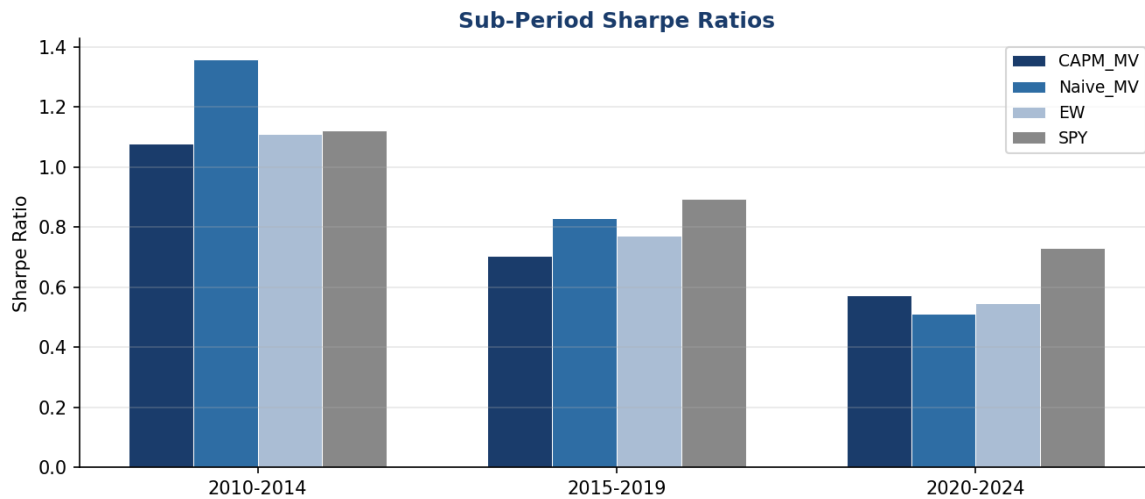


Figure 4. Sub-period Sharpe ratios for all four strategies.

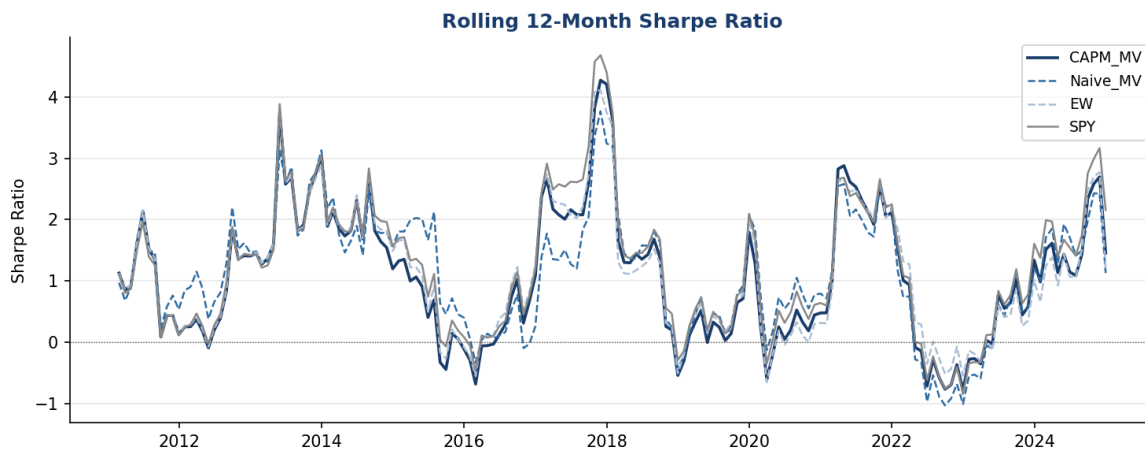


Figure 5. Rolling 12-month Sharpe ratio for all four strategies.

4.5 Historical VaR and Expected Shortfall

Table 5 and Figure 6 report one-month historical 95% VaR and ES for each strategy, using the positive-loss convention. The values are deliberately interpreted as empirical diagnostics rather than parametric tail forecasts: with 178 monthly observations, the 95% ES estimate is driven by roughly the worst nine months.

Table 5. Historical one-month 95% VaR and Expected Shortfall. Values are positive loss percentages.

Strategy	1M VaR 95%	1M ES 95%	Worst Month
CAPM-MV	7.17%	9.99%	−17.17%
Naive MV	7.12%	9.18%	−13.45%
EW	6.05%	9.51%	−16.83%
SPY	6.61%	9.19%	−13.92%

CAPM-MV and Naive MV have similar one-month VaR at approximately 7.2%, but CAPM-MV has a higher ES of 9.99% compared with 9.18% for Naive MV. This does not contradict CAPM-MV's superior maximum drawdown: maximum drawdown is path-dependent, whereas ES averages the

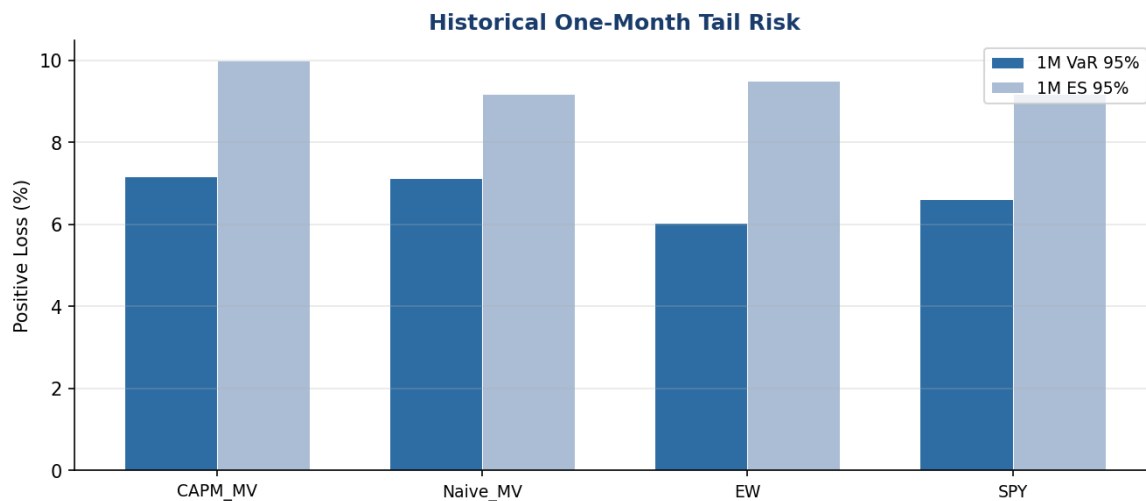


Figure 6. Historical one-month 95% VaR and Expected Shortfall for all four strategies. Values are shown as positive loss percentages.

worst monthly returns. The result shows that CAPM-MV reduced the depth of the 2022 drawdown relative to Naive MV, but its worst individual months, particularly around the COVID crash, were more severe.

4.6 Stress testing and sensitivity analysis

4.6.1 Historical stress scenarios

Table 6 and Figure 7 report cumulative returns across three historical drawdown episodes.

Table 6. Cumulative returns (%) during stress scenarios. All figures net of costs.

Scenario	Period	CAPM-MV	Naive MV	EW	SPY
COVID Crash	Feb–Mar 2020	−26.4	−21.7	−26.1	−22.2
2022 Rate Shock	Jan–Dec 2022	−19.4	−25.2	−12.5	−20.2
2011 Debt Crisis	Aug–Oct 2011	−2.9	−0.6	−2.6	−2.5

The stress windows show different trade-offs. During the COVID window, CAPM-MV performs similarly to Equal Weight and worse than SPY and Naive MV. During the 2022 rate-shock window, it returns −19.4% versus −25.2% for Naive MV, a 5.8 percentage-point advantage; Equal Weight performs best at −12.5%. These differences are consistent with different sector exposures, but a separate attribution analysis would be needed to establish the mechanism. In the shorter 2011 window, all reported losses are smaller.

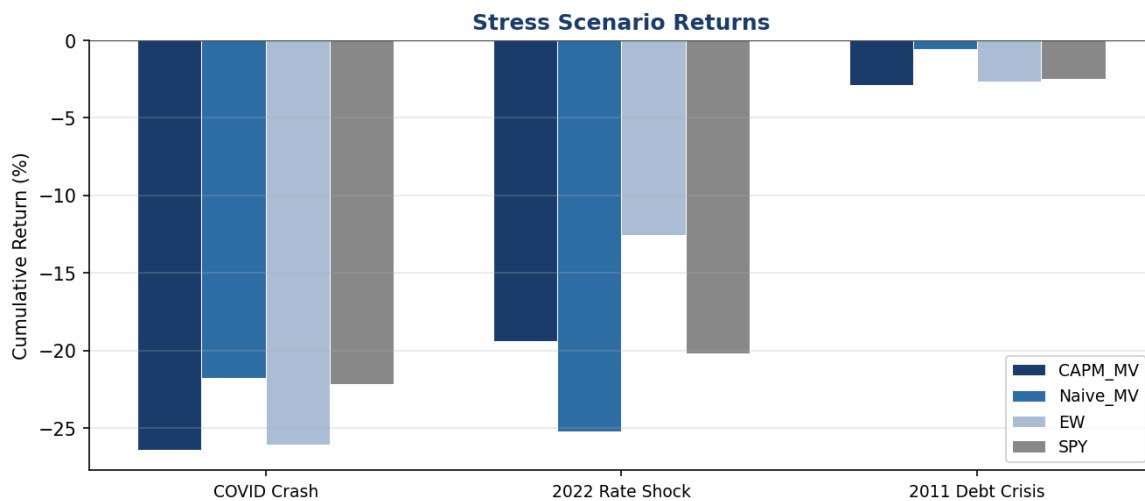


Figure 7. Cumulative returns (%) across the three stress scenarios.

4.6.2 Parameter sensitivity analysis

Figure 8 shows Sharpe ratios as risk aversion λ and the lookback window are varied one-at-a-time around the base case.

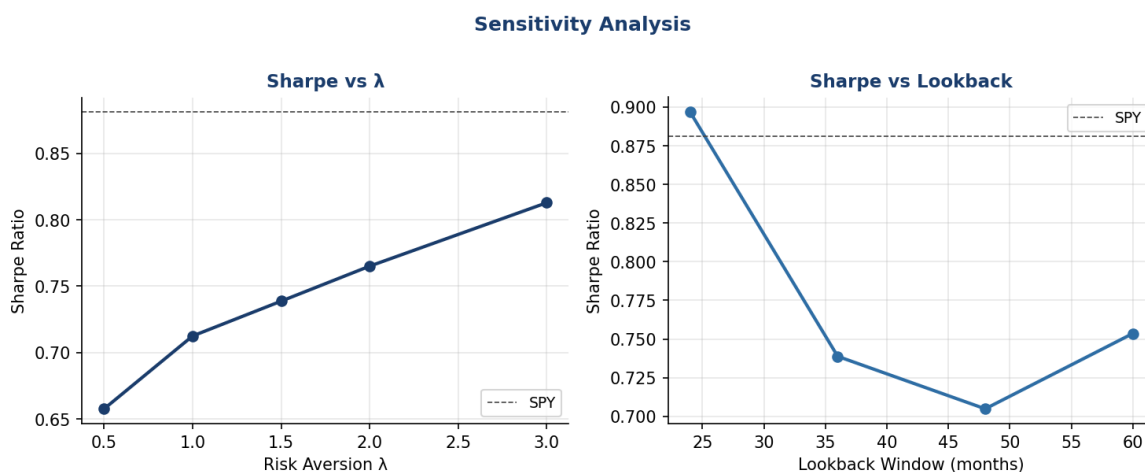


Figure 8. One-at-a-time sensitivity of Sharpe ratio to risk aversion λ (left) and estimation lookback window (right). Dashed line shows SPY Sharpe of 0.881.

The reported ratio increases across the tested λ values. At $\lambda = 3.0$ it reaches 0.813, approaching Naive MV's 0.819, while one-month VaR and ES fall to 5.88% and 8.78%. This demonstrates parameter sensitivity within the evaluated sample; it does not establish that the highest-scoring setting will perform best in a new period.

The 24-month lookback produces a reported ratio of 0.897, above the base 36-month result and SPY's 0.881, with one-month VaR and ES of 6.94% and 9.53%. A shorter window reacts more quickly to changing estimates, but the apparent improvement is selected retrospectively. It should be tested on an independent period or within a nested walk-forward selection process.

Table 7. Sensitivity of Sharpe ratio, maximum drawdown and one-month tail risk to key parameters.

Parameter	Value	Sharpe	Max DD (%)	1M VaR 95%	1M ES 95%
λ sensitivity	0.5	0.658	−29.7	7.33%	10.78%
λ sensitivity	1.0	0.713	−26.8	7.38%	10.34%
λ sensitivity	1.5	0.739	−25.7	7.17%	9.99%
λ sensitivity	2.0	0.765	−26.0	6.90%	9.64%
λ sensitivity	3.0	0.813	−23.4	5.88%	8.78%
Lookback sensitivity	24m	0.897	−26.4	6.94%	9.53%
Lookback sensitivity	36m	0.739	−25.7	7.17%	9.99%
Lookback sensitivity	48m	0.705	−25.6	6.67%	9.83%
Lookback sensitivity	60m	0.754	−24.7	6.34%	9.71%

4.7 Interpretation of the comparison

The full-period results favour Naive MV on return-to-volatility ratio and CAPM-MV on maximum drawdown. Their relative ranking changes in 2020–2024. This is a useful description of the observed trade-off, rather than proof of a general advantage from CAPM regularisation.

The motivation for structuring return estimates is that a sample mean from 36 observations can be imprecise. Under independent monthly observations its standard error is approximately $\sigma_{\text{monthly}}/\sqrt{36}$. The volatility and mean must be expressed over the same horizon. CAPM replaces direct mean estimation with beta and ERP assumptions; it does not eliminate model uncertainty.

Explanations based on technology exposure, beta-estimation lag or the 2022 rotation are hypotheses consistent with the reported weights and returns. A formal attribution study and controlled variants would be required to separate these mechanisms.

5 Limitations and further work

5.1 Modelling and implementation limits

- **Single-factor return model.** CAPM uses market beta and a fixed 5.5% ERP. It omits other return drivers and time variation in the premium. A multifactor comparison, such as the Fama–French three-factor model [6], would test a different set of assumptions.
- **Synthetic ETF histories.** XLC and XLRE are assigned scaled SPY prices before their launches. The early universe is therefore not fully investable as represented, and historical sector exposures are only approximated.
- **Return conventions.** The supplied analysis combines log-return inputs with simple-return compounding for wealth and drawdown. Those conventions should be reconciled before treating the reported wealth paths as an exact investment simulation. Performance ratios labelled Sharpe also omit the risk-free deduction.

- **Restricted universe.** All eleven assets are US equity sectors, limiting diversification across asset classes and markets. Results should not be generalised to a broader opportunity set.
- **Tail risk.** Variance does not describe all aspects of skewed or heavy-tailed returns. Historical 95% ES is based on roughly nine of the 178 monthly observations and is sensitive to individual stress months.
- **Retrospective tuning.** Lookback and risk-aversion settings are compared on the same history. Selecting the best result from that comparison can overstate future performance.

5.2 Further work

First reconcile the return conventions and repeat the analysis on an investable universe. Then evaluate the 24-month lookback and $\lambda = 3.0$ alternatives using an independent validation design. The latter reports a ratio of 0.813 and maximum drawdown of -23.4% in the current sensitivity exercise.

Additional comparisons could include a multifactor return model, Black–Litterman views [7], or a CVaR objective that targets tail losses directly. A volatility-based risk overlay could also be tested, with transaction costs and timing rules fixed in advance. None of these extensions is assumed to improve performance without evaluation.

6 Conclusion

The project implements a rolling CAPM-informed allocation framework and compares it with historical-mean optimisation, equal weighting and SPY. Its main contribution is a transparent comparison of return-input choices under common constraints.

CAPM-MV's reported ratio of 0.739 is below Naive MV's 0.819 and SPY's 0.881. It nevertheless has a less severe maximum drawdown than Naive MV (-25.7% versus -30.7%) and a higher Calmar ratio (0.447 versus 0.399). Its one-month 95% VaR of 7.17% and ES of 9.99% show why drawdown and monthly tail losses should be assessed separately.

The 24-month lookback and higher risk-aversion settings are candidates for further testing. Correcting the return-convention and investability limitations, then validating parameter choices on independent data, would strengthen the interpretation of the results.

References

- [1] Markowitz, H. (1952). Portfolio Selection. *Journal of Finance*, 7(1), 77–91.
- [2] Sharpe, W. F. (1964). Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk. *Journal of Finance*, 19(3), 425–442.
- [3] Ledoit, O. and Wolf, M. (2004). Honey, I Shrunk the Sample Covariance Matrix. *Journal of Portfolio Management*, 30(4), 110–119.
- [4] Ledoit, O. and Wolf, M. (2004). A Well-Conditioned Estimator for Large-Dimensional Covariance Matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.
- [5] Damodaran, A. (2024). Equity Risk Premiums: Determinants, Estimation and Implications. *NYU Stern Working Paper*. Available at: <https://pages.stern.nyu.edu/~adamodar>
- [6] Fama, E. F. and French, K. R. (1993). Common Risk Factors in the Returns on Stocks and Bonds. *Journal of Financial Economics*, 33(1), 3–56.
- [7] Black, F. and Litterman, R. (1992). Global Portfolio Optimization. *Financial Analysts Journal*, 48(5), 28–43.
- [8] DeMiguel, V., Garlappi, L. and Uppal, R. (2009). Optimal versus Naive Diversification: How Inefficient is the $1/N$ Portfolio Strategy? *Review of Financial Studies*, 22(5), 1915–1953.
- [9] scikit-learn developers. *LedoitWolf*. Estimator documentation. <https://scikit-learn.org/stable/modules/generated/sklearn.covariance.LedoitWolf.html>